NETWORK TRAFFIC ANALYSIS USING WIRESHARK

Beginner Cybersecurity Project Report

Project type	Hands-on network analysis
Tool	Wireshark on Windows with Npcap
Scope	DNS queries/responses, TCP packets, and visible TLS handshake messages
Capture interface	Wi-Fi

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This report documents a beginner-level exercise using Wireshark to observe network traffic and learn how DNS, TCP, and TLS appear in packet captures. Replace the placeholders and confirm all observations before publishing.

1. Abstract

This project demonstrates basic network traffic analysis using Wireshark. Traffic was captured from the Wi-Fi interface on a Windows laptop and examined with display filters. The `dns` filter was used to inspect DNS queries and responses, while `tcp` was used to inspect TCP communication, ports, flags, and TLS-related traffic. The exercise reinforces networking fundamentals and introduces a repeatable workflow for documenting packet-level observations.

2. Objectives

• Understand the purpose of packet capture and Wireshark.

• Identify DNS queries and responses and recognise common DNS record types.

• Inspect TCP packets, source/destination addresses, ports, and TCP flags.

• Recognise TLS handshake messages and understand that application traffic is encrypted.

• Document findings clearly and safely for learning and portfolio use.

3. Tools and Environment

Tool / component	Purpose
Wireshark	Captures and decodes network packets for analysis.
Npcap	Enables live packet capture on supported Windows interfaces.
Windows laptop + Wi-Fi	Environment used for the capture.

4. Methodology

4.1 Start capture

Wireshark was opened and the active Wi-Fi interface was selected to capture live network traffic.

4.2 Generate normal traffic

Normal browsing activity was used to produce packets for observation.

4.3 Analyse DNS

The display filter `dns` was applied to isolate DNS packets. Queries and responses for domain names were visible in the packet list.

4.4 Analyse TCP

The display filter `tcp` was applied to show TCP packets. The packet list showed addresses, ports, acknowledgements, and TCP flags.

4.5 Inspect TLS

TLS-related rows, including Client Hello, Server Hello, and Application Data, were observed in the capture. TLS encrypts application content for protected connections.

5. Filters Used

Display filter	What it shows
`dns`	DNS queries and responses visible in the capture.

`tcp`	TCP packets, including connection setup and data-transfer acknowledgements.

6. Observations

• DNS packets included standard queries and query responses for domain names visible during normal network activity.

• DNS rows included different record types such as A, AAAA, and HTTPS in the observed capture.

• TCP packets showed source and destination addresses and port numbers; port 443 appeared in traffic associated with HTTPS.

• A selected TCP packet displayed SYN, ACK flags. This is consistent with the server side of a TCP three-way handshake, when paired with the corresponding earlier SYN packet.

• TLS-related packets included handshake messages and Application Data, illustrating encrypted application communication.

Note: These observations are based on the screenshots reviewed during the exercise. Confirm the final capture and update any details if your saved file differs.

7. Results and Learning Outcomes

The exercise provided practical exposure to packet capture and display filtering. It connected networking concepts—DNS resolution, TCP ports and flags, and TLS—to packets observed in Wireshark. The workflow can be repeated in an authorised lab to investigate other protocols.

• Applied Wireshark display filters successfully.

• Distinguished DNS queries from DNS responses.

• Identified common TCP fields and flags.

• Recognised TLS handshake and encrypted application traffic.

• Practised saving evidence and documenting technical findings.

8. Limitations

This is an introductory exercise, not a full security audit or proof that the network is safe or compromised. A packet capture shows only traffic visible to the capture interface during the selected period. Encrypted TLS application data generally cannot be read as plain text from the capture alone. No attack or malicious activity was claimed or tested.

9. Ethical and Privacy Considerations

Capture only traffic on devices and networks you own or are authorised to inspect. Packet captures may contain IP addresses, hostnames, metadata, and other sensitive details. Before publishing the project, upload only a sanitised report and selected screenshots. Do not upload the raw .pcapng file publicly unless you have reviewed and removed sensitive data.

10. Conclusion

This beginner project successfully introduced the use of Wireshark for network traffic analysis. DNS filtering helped reveal name-resolution activity, while TCP filtering exposed transport-layer communication and flags. Observed TLS messages demonstrated how secure application traffic appears in a packet capture. The project provides a foundation for further study of network protocols, SOC monitoring, and incident investigation.